

Extra homework 4

1. Rearrange the terms in the alternating harmonic series such that its sum is $s \in \overline{\mathbb{R}}$.
2. Let (F_n) be the Fibonacci sequence, with $F_0 = F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$. Study the convergence and find the value of the series $\sum_{n=0}^{\infty} F_n x^n$.
3. Let C_n be the number of full binary trees with $n + 1$ leaves (known as the Catalan number).
 - (a) Find a recurrence relation for C_n .
 - (b) Considering the generating function $f(x) = \sum_{n=0}^{\infty} C_n x^n$, prove that $C_n = \frac{1}{n+1} \binom{2n}{n}$.

These questions are extra. You will get bonus points for solving them.